

AMD DEVELOPER HACKATHON · ACT II

Nova

A full-stack AI agent platform that runs its inference on **AMD Instinct** compute.

Track 3 · Unicorn (flagship)

Track 1 · General-Purpose Agent

Gemma side-prize ready

THE OPPORTUNITY

Agent platforms are model-locked. Nova is not.

Teams want capable agents without being trapped by one vendor's API, price, or hardware. Nova is a complete agent workspace — planning, tools, a live "Agent's Computer", sandboxed browser & editor — with a **config-driven model layer** that runs on whatever compute you choose.

- Swap providers with a config entry — no code changes.
- First-class support for **AMD Instinct** compute, managed or bare-metal.
- Token usage tracked end-to-end, so efficiency is measurable.

Nova is a shipped product, not a demo

Agent workspace

Multi-step planning, tool use, an "Agent's Computer" with live terminal, editor, and sandboxed browser preview.

Provider system

Any OpenAI-compatible, Ollama, vLLM, Anthropic, Gemini endpoint — added via UI or config, hot-reloaded.

Ops hardened

PM2 full-stack, a self-healing watchdog, regression tests on every fix from live audits.

Token telemetry

Per-model input/output token accounting surfaced in the workspace.

Two AMD compute paths, one-click in Nova

Fireworks AI · managed

Inference served on **AMD Instinct MI300X**. Added via the "Add Fireworks model (AMD MI300X)" preset. Powers the Track 1 agent.

AMD Developer Cloud · bare-metal

Nova serves its **own** inference with **vLLM on ROCm** on an AMD Instinct instance — `scripts/amd-serve-vllm.sh`.

Adding AMD compute is configuration, not code: Nova's model layer already treats every provider as a `use` + `base_url` + `api_key` entry.

ARCHITECTURE

Config-driven providers → AMD compute

- **UI / API** → runtime model store (`runtime_models.yaml`), env-driven (`$FIREWORKS_API_KEY` resolved at load, never persisted).
- **Model factory** instantiates the LangChain provider from `use:` — Fireworks via `ChatOpenAI` , AMD Instinct via the native `VllmChatModel` .
- **Telemetry**: every run's provider is tagged AMD-backed; an `AMD` badge appears in the UI.
- **Signal**: `GET /api/models/amd-usage` reports AMD usage as machine-readable JSON for pre-screening.

DEMONSTRABLE — AND HONEST

AMD usage you can verify

```
curl /api/models/amd-usage →
```

```
{ "amd_backed": true, "count": 2, "summary": "Nova is running on AMD compute: 2 AMD-backed model(s)..." }
```

- Only verifiably AMD-hosted endpoints (Fireworks) are auto-claimed.
- Self-hosted vLLM must opt in explicitly — a bare vLLM server is never assumed to be AMD.
- The badge never overstates where inference actually ran.

One product, all three tracks

Track 3 · Unicorn **(flagship)**

Nova as an original, AMD-powered agent platform. Repo + live demo + this deck.

Track 1 · General agent

Lean Fireworks batch harness, token-efficient by design.
`make hackathon-track1`.

Gemma side prize

Gemma is a default in the AMD presets (Fireworks + vLLM/ROCm).

Track 2 · Video captioning

Reuses Nova's vision layer — planned stretch on the same provider system.

WHY IT MATTERS

Market potential

Every team adopting agents faces the same buy: capable, controllable, cost-predictable inference. Nova decouples the agent product from the model vendor — and makes **AMD Instinct** a first-class, one-click backend. That is a credible on-ramp for AMD compute into the fast-growing agent tooling market.

- **For teams:** vendor-flexible agents, measurable token cost.
- **For AMD:** a polished product that showcases MI300X + ROCm to agent builders.

NOVA

Agents on AMD.

Complete product · genuine AMD
compute · verifiable usage.

- Repo: `github.com/<org>/nova`
- Live demo: `<hosted-url>`
- AMD setup: `docs/AMD_INTEGRATION.md` · usage: `/api/models/amd-usage`